Faiyaj Rahman

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Education

University of Michigan – Ann Arbor **Ann Arbor, MI**

B.S.E. in Computer Engineering | GPA: 3.6/4.0 · Dean's List Expected Graduation: May 2029

- **Coursework:** Data Structures & Algorithms, Computer Architecture, Object Oriented Programming, Discrete Math, Computing Systems, Linear Algebra, Multivariable & Vector Calculus

Honors

Jane Street FOCUS | *Selected from 1000+ applicants for quant trading and SWE program* May 2026

Work Experience

Bosch **Farmington Hills, MI**

Software Engineering Intern May 2026 – Aug. 2026

- Deployed a Flask operations platform to AWS and scripted multi-tenant KPIs and analytics scaling to **5000+** devices.
- Designed a Python tool to convert CAN `.asc` logs to `MF4` with DBC channel mapping, saving **20+** engineer-hours monthly.
- Shipped a 3-agent pipeline, cleaning and tagging internal documents for a master agent, cutting new-hire escalations **40%**.
- Engineered a multi-threaded GUI tool used by **100+** developers to build Python executables, saving **$15K** in licenses.

Hikmah Hackers **Warren, MI**

Co-Founder & Lead Instructor Sep. 2026 – Present

- Co-founded a coding and mentorship program for underserved middle and high school students at a STEM school, teaching programming fundamentals through problem sets & hands-on mini-projects, and giving students tailored career guidance.
- Designing a curriculum spanning CS topics, game/web dev, and AI scripting, featuring hackathons offering cash prizes.

Saf · Darul Uloom Michigan **Warren, MI**

Software Developer Nov. 2025 – Present

- Launched a school-management beta on Postgres, an Express REST API, and React featuring role-based access control.
- Implemented assignments, grading, and attendance, replacing paper grades for **20+** educators in a **3**-month live pilot.
- Driving AI integration on a 3-engineer team, scoping LLM report card narratives & role-scoped natural language querying.

MRacing FSAE **Ann Arbor, MI**

Software Engineer Aug. 2025 – Present

- Built a C++ Raspberry Pi dashboard decoding **1000+** CAN messages/sec into driver analytics & fault states over **3** pages.
- Collaborated with engineers to integrate vehicle telemetry, CAN diagnostics, and fault checking into the racing dashboard.
- Wrote C firmware for an infrared lap trigger, edge-detecting photosensor voltage drops on beam breaks to timestamp laps.

University of Michigan Electric Vehicle Center **Ann Arbor, MI**

Undergraduate Research Software Developer Sep. 2025 – May 2026

- Developed open-source Python UI software for an EV charging module, contributing to **$10K** in net savings for an OEM.
- Designed a Simulink finite-state machine for system-safety logic and charging protocols, translating models into Python.
- Integrated state logic into the software pipeline, coordinating state transitions, charging sequences, and safety conditions.

Projects

Stock Market Simulator & PQ Libraries | *C++, Linux, Templated design, Git* May. 2026 – Oct. 2026

- Developed a C++ exchange simulator matching orders with **price–time priority**, partial fills, and multi-stock execution.
- Implemented **3** priority queues (binary, sorting and pairing heaps) to test insertion, removal, and matching performance.
- Built **4** reporting modes for live trades, running-median prices, trader PnL, and optimal historical buy/sell opportunities.

Low-Level Modulator | *Verilog, Assembly, FPGA, Digital Logic* Jan. 2026 – Apr. 2026

- Engineered an **18**-instruction processor in Verilog on a DE2-115 FPGA, driving a synthesizer with **40** assembly modules.
- Designed an integer-math DSP pipeline synthesizing **32-bit** audio from **3** wavetables with **5** modulation modes at **30 FPS**.
- Wrote instructions (arithmetic operations and control logic) for processor execution with FPGA-based audio generation.

Technical Skills

Languages: Python, C, C++, JavaScript, TypeScript, SQL, Lua, Julia, Bash, Assembly, Verilog, MATLAB
Tools/Frameworks: Claude Code/Cursor Skills, React, Node/Express, Tailwind, PySide6, Flask, GoogleTest, PostgreSQL, Supabase, AWS, Docker, Kubernetes, Linux, GitLab, CAN, Raspberry Pi, Simulink
Practices: Agile/Scrum, Git/GitHub, CI/CD, code review, unit testing, LLM agents
Interests: Traveling, Gaming, Soccer, Weightlifting, Cooking, Studying Islam, Watching Movies